Check for updates

Drug Testing and Analysis

WILEY

RESEARCH ARTICLE OPEN ACCESS

Assessment of Abuse Potential-Related Effects of Oral Dried Kratom Leaf Powder in Healthy Normal Participants Following Single and Multiple Daily Doses

Marion Coe[1] | Marilyn Huestis[2] | Daniel Wang[1] | Thomas Hudzik[3] | John Bothmer[4] | Jack Henningfield[1,5]

[1]Pinney Associates Inc., Bethesda, Maryland, USA | [2]Institute of Emerging Health Professions, Thomas Jefferson University, Philadelphia, Pennsylvania, USA | [3]ALA BioPharma Consulting, Monroe, New Jersey, USA | [4]JB PharmaConsult, Heerlen, the Netherlands | [5]Johns Hopkins University School of Medicine, Department of Psychiatry and Behavioral Sciences, Baltimore, Maryland, USA

Correspondence: Marion Coe (mcoe@pinneyassociates.com)

Received: 3 February 2026 | **Revised:** 24 April 2026 | **Accepted:** 30 April 2026

Keywords: abuse liability | double-blind | kratom | mitragynine | placebo-controlled | randomized | safety

ABSTRACT

Kratom is increasingly used in the United States for energy and self-management of pain and mood. While its active alkaloid, mitragynine, has partial μ-opioid receptor agonist activity, clinical data on its abuse potential remain limited. This study evaluated abuse-related outcomes following single and multiple doses of dried kratom leaf powder in healthy kratom-naive adults in a randomized, double-blind, placebo-controlled study. Healthy participants ($n = 116$) received a single oral dose or 15 consecutive daily doses of Mitra-Leaf kratom capsules (4 cohorts: 500–4000 mg dried leaf powder; equivalent to 6.65–53.2 mg mitragynine; $n = 12$–13 per cohort, $n = 49$ total) or placebo ($n = 67$). Subjective effects were assessed using the Drug Effects Questionnaire (DEQ). Opioid withdrawal symptoms were evaluated using the Subjective Opioid Withdrawal Scale (SOWS) and Clinical Opiate Withdrawal Scale (COWS). Abuse-related treatment-emergent adverse events (ARTEAEs) were recorded. DEQ responses were dose-related, except between the middle two dose levels with statistically significant increases in maximum effect and area-under-the-curve compared to placebo observed only at the highest single dose. DEQ responses decreased following repeated dosing. No participants met the threshold for clinically meaningful opioid withdrawal based on COWS or SOWS. ARTEAEs occurred more frequently at higher doses but were generally mild and resolved without sequelae. No serious adverse events or deaths were reported. Oral administration of kratom at tested doses was well tolerated and associated with modest, dose-related subjective effects, minimal withdrawal symptoms, and low ARTEAE rates. Findings support further research on kratom's safety and therapeutic potential, while emphasizing the importance of dose control and product standardization.

1 | Introduction

Kratom is a botanical supplement derived from the leaves of the *Mitragyna speciosa* (Korth.) tree native to Southeast Asia that has been used there for centuries as traditional medicine, and now more recently in the United States and other western countries for various health-related reasons, including to alleviate pain, reduce fatigue, boost mood, and anecdotally to manage opioid withdrawal [1–4]. As kratom consumption in the United States has become more widespread, it has gained significant attention from the public, regulatory authorities, and researchers—both for its potential benefits and due to questions pertaining to its safety and potential for abuse [3, 5], but well-conducted clinical studies examining both aspects of kratom use are needed.

The primary active alkaloid present in kratom is mitragynine, which is likely responsible for many of its effects. In animal drug discrimination studies, mitragynine most robustly generalizes

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2026 The Author(s). *Drug Testing and Analysis* published by John Wiley & Sons Ltd.

to alpha-adrenergic acting lofexidine [6] and it has agonistic effects at adrenergic, serotonergic, D2-dopaminergic, and adenosine receptors [7–11]. Mitragynine is a partial agonist of the human μ-opioid receptor and is a competitive antagonist at κ- and δ-opioid receptors. It has lower potency and affinity for the μ-opioid receptor than prototypical opioids such as morphine and low beta arrestin ("biased G-protein") effects, which likely contribute to its lower respiratory depressant effects and abuse potential [12, 13].

Although mitragynine is the most abundant naturally occurring alkaloid in kratom leaves, there are more than 40 additional alkaloids with varying levels of types of activity that may also contribute to the effects experienced by consumers [14–17]. 7-Hydroxymitragynine, another alkaloid found in significantly lower amounts in kratom, has much stronger morphine opioid receptor binding and pharmacologic effect than most other kratom alkaloids and metabolites, but it is more appropriately considered a mitragynine metabolite than kratom constituent [8, 18, 19].

Taken together, the overall pharmacology and mechanisms of action of mitragynine and kratom generally help increase the understanding of the safety profile and diverse reasons for kratom use reported by consumers. For example, whereas mitragynine (5.6 and 10 mg/kg, intravenous [i.v.]) exhibits respiratory stimulating activity in rats, in the same study morphine produced substantial depression [20]. In a study involving oral administration of mitragynine and oxycodone (by gastric lavage) following an FDA recommended protocol for assessing respiratory effects by blood gas assessment, a broad range of mitragynine doses produced no significant reductions in blood oxygen or life threatening effects at any dose, whereas oxycodone produced profound blood oxygen reductions and deaths at the two highest doses recommended by FDA [21].

Analogously, whereas morphine and heroin maintained robust self-administration in the i.v. rat model, mitragynine was similar to saline with very low rates of self-administration [22, 23]. Due to this unique pharmacology, it is unsurprising that kratom and mitragynine-containing products are being consumed for pain relief, self-management of opioid and other drug withdrawal, and management of substance use disorders [2]. Reasons for use also include improved quality of life, increased productivity, and self-management of stress, mood, and sleep issues [3, 24–26].

The second most frequently reported reason for kratom consumption, after self-management of pain, found in a US nationally representative survey, was to boost energy. This is consistent with the findings of other surveys in which people reported use of kratom in place of or in addition to caffeinated beverages to boost energy and for its useful "instrumental" effects analogous to commonly reported reasons for caffeinated beverage consumption [25, 26]. This finding is not surprising given the botany of *Mitragyna speciosa*, which belongs to the Rubiaceae family that includes coffee trees [27, 28].

It is important to understand the abuse potential of this substance given its use for a wide array of reasons. A complete picture of abuse potential relies on pharmacology, animal self-administration models, clinical studies, and real-world evidence. An assessment of the abuse potential of kratom according to the eight factors of the Controlled Substances Act was published [9] and concluded that kratom did not pose a significant public health risk warranting Controlled Substance Act control, and to the contrary, it was in the interest of public health that kratom not be banned from licit sales and possession. This was also the conclusion of the Assistant Secretary of Health of the US Department of Health and Human Services [29].

Physical dependence and withdrawal associated with kratom are also generally milder and qualitatively distinct from that produced by opioids in animal and human studies [30]. In humans, there are several case reports of withdrawal symptoms due to cessation of kratom [31, 32], and online surveys of kratom users indicate that anywhere between 10% and 20% of respondents report possible kratom-related withdrawal symptoms such as craving, fatigue, irritability, and low energy [2, 33–35]. Importantly, these symptoms were typically rated as mild and not significantly impacting daily life [36, 37]. For comparison, amongst people who regularly consume caffeine, approximately 8% meet diagnostic criteria for caffeine use disorder [38] and between 35.9% and 46.6% experience some form of withdrawal upon cessation [39, 40].

The reported effects of kratom vary widely and depend on dose, frequency and history of use, and alkaloid content of the product used [41]; therefore, carefully controlled investigations of standardized kratom products are needed. The objective of this study was to evaluate potential adverse effects relied upon by the US FDA for assessing abuse potential and physical dependence and instruments validated for opioid withdrawal assessment in a single-dose exposure condition and a "chronic" condition of 15 consecutive daily oral doses of dried kratom leaf powder in healthy adults.

2 | Methods

2.1 | Ethical Practices

This study was conducted in accordance with International Conference on Harmonisation Good Clinical Practice (ICH GCP) guidelines and ethical principles that have their origin in the Declaration of Helsinki. Protocols were approved by an Independent Ethics Committee and Health Canada before eligibility screening. Written informed consent was obtained from each participant before any study-related procedures were performed.

2.2 | Study Participants

Healthy, nonsmoking adults (aged 18–55 years) with a body mass index between 18.5 and 29.9 kg/m^2 inclusive were enrolled. Female participants were of non-childbearing potential or were non-pregnant and non-lactating. Appropriate birth control was utilized by all participants throughout the study, with females of childbearing potential testing negative on pregnancy tests and not breastfeeding and having no intention to get pregnant or breastfeed throughout the study. Participants were required to be either naïve to kratom or to have not reported using it for

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

a minimum of 1 year. The presence of gastrointestinal, hepatic, renal, cardiovascular, respiratory, autoimmune, endocrine, neurological, or active psychiatric diseases or positive serology results for HIV, HBsAg, or HCV or a history of cancer were exclusionary. Also exclusionary were an acute illness within 2 weeks of the single dose, a positive urine drug test at screening, a history of drug or alcohol abuse in the past 12 months, a Leeds Dependence Questionnaire score ≥ 21, or consumption of ≥ 2 alcoholic beverages a day. Hepatic or renal dysfunction with alanine aminotransferase (ALT) or aspartate aminotransferase (AST) ≥ 2× upper limit of normal (ULN) and serum creatinine or blood urea nitrogen (BUN) ≥ 1.5× ULN were evaluated by the principal investigator for inclusion or exclusion. Participation in another research study or blood donation within 4 weeks prior to single dosing was prohibited. Participant recruitment and stratification was employed to balance the ratio of sex between active and placebo groups. Medications that induced enzymes were not permitted for 14 days or 7 half-lives of Day 0 or within 7 days if not an enzyme inducer. Participants were required to appear at the study site for 31 visits during the active portion of the study for dosing and other assessments.

2.3 | Study Design

The present study was conducted between date and date at a single site (NutraSource, Guelf, Ontario, Canada). There were two phases in the study: a single-dose (SD) phase and a multiple-dose (MD) phase. Each phase investigated the safety, tolerability, pharmacokinetics, and central nervous system effects of encapsulated Mitra-Leaf powder (NP Pharma). Pharmacokinetic results are published elsewhere [42]. All visits and dosing occurred on site. Participants attended 1 screening visit to assess eligibility, which occurred up to 28 days prior to enrollment, and 1 training visit, which occurred up to 14 days prior to dosing day and familiarized the participant with the study questionnaires.

Single-dose phase: The SD phase was double-blind, randomized, and placebo controlled. Four groups of 12–13 healthy participants were enrolled and received a single oral dose of 6.65 mg, 13.3 mg, 26.6 mg, and 53.2 mg mitragynine contained in 500, 1000, 2000, and 4000 mg kratom raw leaf powder in 1, 2, 4, or 8 capsules, respectively, with associated placebo groups ($n = 67$). The placebo group is larger than the active group because this manuscript provides data from one formulation of kratom tested in a larger clinical study including other formulations not described here; thus, there were matching placebo groups for all formulations. Matched placebo capsules contained inactive ingredients and were identical in fill volume, weight, and appearance to the active ingredient in the investigational product. See Table 1 for detailed sample sizes. Administration of each successive dose depended on the safety and tolerability of previous doses.

Multiple-dose phase: The MD phase was double-blind, randomized, and placebo controlled, utilizing the same participants as the SD phase. The MD phase initiated 10 days after dosing in the SD phase, and pharmacokinetic testing was conducted to ensure complete washout of investigational product had occurred. Participants then received a daily single oral dose of 500, 1000, 2000, and 4000 mg kratom raw leaf for 15 consecutive days. The associated placebo groups ($n = 67$) utilized matching capsule appearance, weight, and number. Participants were not re-randomized; thus, they received multiple administrations of the same dose administered to them in the SD phase.

2.4 | Investigational Product Formulation and Administration

Kratom capsules were labeled according to ICH GCP and Health Canada guidelines and stored at room temperature (20–25 °C) and relative humidity ≤ 65%. The product contains between 0.8 and 1.4% mitragynine weight/leaf powder weight and met US Pharmacopeia (USP) standards for impurities (i.e., no impurities at or above levels accepted for dietary supplements). Each capsule consisted of 500 mg kratom leaf powder containing 6.65 mg of mitragynine. Matched placebo capsules contained inactive ingredients and were identical in fill volume, weight, and appearance as the active ingredient in the investigational product.

2.5 | Outcome Measures

2.5.1 | Subjective Drug Effect

Participants were queried on a 100 mm visual analog scale asking "Do you feel a drug effect right now?" This was asked on Day 0 (after SD administration) and Day 24 (after the last [i.e., 15th] dose of the MD phase) at the following time points: pre dose/baseline, and at 15-min intervals up to 3 h post-dosing, then hourly to 12 h post-dosing (i.e., 15 min, 30 min, 45 min, 1 h, 1 h 15 min, 1 h 30 min, 1 h 45 min, 2 h, 2 h 15 min, 2 h 30 min, 2 h 45 min, 3 h, 4 h, 5 h, 6 h, 7 h, 8 h, 9 h, 10 h, 11 h, and 12 h).

2.5.2 | SOWS and COWS

The SOWS is a self-administered scale for grading opioid withdrawal symptoms. It contains 16 symptoms that are rated on a scale of 0 (not at all) to 4 (extremely), with a total score range of 0 to 64. It was administered on Day 7 (7 days after SD), prior to dosing and 4, 8, and 12 h after dosing on the last day of MD phase (i.e., after 15 days of consecutive dosing), and 4 times per day for 3 days after the last dose of the MD phase. The COWS consists of 11 opioid-withdrawal signs rated by a trained observer on a scale from 0 to 4 (total scale range, 0–44, where 5–12 represents mild symptoms). It was administered on Day 7 (7 days after SD), prior to dosing and 12 h after dosing on Day 24 (last day of MD phase), and once per day for 3 days after the last dose of the MD phase.

Adverse Events: Adverse events (AEs) were not specifically queried during the study. When spontaneously reported, relatedness, time since last dose, severity, and outcome were all recorded. An AE was defined as any undesirable symptom or occurrence experienced by a participant during a clinical study, including worsening of a pre-existing condition recorded at Screening. Severity was characterized according to the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use Guidelines. Briefly, an AE was considered mild if it met the following criteria: transient or easily tolerated symptoms, minimal discomfort,

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Participant demographics.

	Single-dose phase					Multiple-dose phase				
	Mitragynine/dried kratom leaf powder (mg, PO)									
	0/0	**6.65/500**	**13.3/1000**	**26.6/2000**	**53.2/4000**	**0/0**	**6.65/500**	**13.3/1000**	**26.6/2000**	**53.2/4000**
Participants (*n*)	67	12	12	12	13	60	11	11	10	12
Sex										
Female	33	6	4	4	7	28	6	4	3	6
Male	34	6	8	8	6	32	5	7	7	6
Race										
American Indian	3	0	0	0	2	2	0	0	0	2
Asian	10	2	5	1	0	9	2	5	0	0
Black	12	0	0	4	1	10	0	0	3	1
Unknown	2	0	0	0	1	1	0	0	0	1
White	40	10	7	7	9	38	9	6	7	8
Ethnicity										
Hispanic	6	2	1	0	3	4	2	1	0	3
Not Hispanic	61	10	11	12	10	56	9	10	10	9
Age (mean, years)										
Mean	32.7	30.8	32.9	35.8	33.2	33.4	31.3	33.9	37.3	33.9
(SD)	(9.9)	(9.3)	(9.4)	(11.4)	(9.6)	(10.0)	(9.5)	(9.2)	(12.1)	(9.7)
BMI (mean, kg/m^2)										
Mean	25.4	23.1	25.3	26.2	23.4	25.5	22.8	25.4	26.1	23.8
(SD)	(2.7)	(2.7)	(2.6)	(3.4)	(2.9)	(2.7)	(2.6)	(2.7)	(3.7)	(2.8)

Note: Numbers here include all participants who received at least a single dose of investigational product (active or placebo) during each study phase. Participants completed both phases sequentially.
Abbreviations: BMI, body mass index; PO, *per os* (oral); SD, standard deviation.

19427611, 2026, 7, Downloaded from https://analyticalscience.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

no interference with daily activities, and no or minimal medical intervention required. It was considered moderate if it met any of the following criteria: greater discomfort, interferes with usual daily activities, or may require medical intervention or treatment. A severe AE was characterized by any of the following: incapacitated or prevented normal daily activities, marked symptoms or significant clinical concern, or required intensive treatment or hospitalization.

If an AE's onset was on or after the first dose or the pre-existing medical condition was aggravated in severity or frequency following study intervention, it was treated as treatment-emergent adverse event (TEAE). The frequency (number of events and number of participants) and incidence (percentage of participants) of each TEAE in each active formulation and pooled placebo groups were reported if they occurred in more than 1 participant. All AEs were followed to resolution. A complete summary of AEs is available [43], but the focus of this publication is to discuss potentially abuse-related AEs; therefore, those are presented herein. Abuse-related AE terms were selected according to the Recommended List of Abuse-related AEs to Assess Abuse Potential, as presented at the Cross-Company Abuse Liability Counsel 2023 Meeting [44]. Withdrawal-related TEAEs were characterized according to definitions in Kosten, Baxter [45].

2.6 | Data Analysis

2.6.1 | Safety Analyses

These were performed for all participants who received at least one dose of study intervention in both SD and MD phases. These analyses considered AEs, which were monitored throughout the study, and which included any undesirable symptom or occurrence experienced by a participant during the study. Only AEs that were treatment-emergent and suspected to be related to the study treatment (excluding pre-existing conditions) were considered for these analyses (see supplemental materials for TEAEs regardless of suspected relation to study treatment). Analyses also separately considered AEs on the recommended list of abuse-related AEs to assess abuse potential as presented at the Cross-Company Abuse Liability Counsel (CCALC; see Supplemental Table 1 [44]). The number and percentage of participants who experienced an AE are presented for each group.

2.6.2 | DEQ

Area-under-the-curve (AUC) and maximum effect (Emax) for DEQ were compared. Because scores were not normally distributed (right-skewed), Mann–Whitney *U* tests were performed. Each dose was compared to pooled placebo, and scores from the SD phase were compared to the MD phase within each dose. Because a large number of tests were performed (four doses across two phases [SD and MD] compared to placebo for 8 tests, and four doses compared to themselves across phases for an additional four tests, or 12 tests total for AUC and 12 tests total for Emax), Bonferroni corrections were made so that the alpha levels for these tests were $0.05/12 \approx 0.004$. As a sensitivity test, the same tests were also considered without Bonferroni correction.

2.6.3 | COWS and SOWS

Mean COWS and SOWS total scores were compared to placebo for each dose. Because COWS and SOWS total scores were not normally distributed (right-skewed), Mann–Whitney *U* tests were performed. Because a large number of tests were performed (four doses across 6 time points for a total of 24 tests for COWS and four doses across 6 time points for a total of 24 tests for SOWS), Bonferroni corrections were made so that the alpha levels for these tests were $0.05/24 \approx 0.002$ for COWS and for SOWS. As a sensitivity test, the same tests were also considered without Bonferroni correction.

3 | Results

3.1 | Participants

A total of 116 participants were enrolled in the study and received at least a single dose of investigational product: 49 received active ingredient in the investigational product (12 participants each received the three lowest doses, 13 received the highest dose) and 67 received placebo. Participant demographics are shown in Table 1.

3.2 | Subjective Drug Effects

Overall, DEQ scores increased with increasing dose (except between the middle two dose levels). The data show that mean Emax and AUC DEQ values after the last day of the MD phase are lower than the DEQ values after a single dose. There were three participants who withdrew from the study before the last day of the MD phase that had above-average DEQ scores during the SD phase. This may have in part contributed to the observed reductions from SD to the MD phase in mean DEQ Emax and AUC. However, for most participants with DEQ results in both SD and MD phases, the DEQ maximum value and AUC were still higher after dosing in the SD phase. Importantly, because three participants with above-average DEQ responses withdrew prior to completion of the MD phase, the interpretation of reduced subjective effects over time must be limited. DEQ findings are shown in Table 2.

Mann–Whitney *U* tests indicated that there was no detectable difference in AUC at any dose or phase compared to placebo, nor at any dose between phases, when the Bonferroni correction was applied ($56.5 \leq Us \leq 651.5$; $ps > 0.004$). However, AUC was greater in the highest dose group (53.2 mg, mitragynine) compared to placebo for SD phase when the Bonferroni correction is *not* considered (84.3 vs. 16.4; $U = 651.5$; $p = 0.005$).

Mann–Whitney *U* tests indicated that Emax was statistically significantly greater in the highest dose group (53.2 mg, mitragynine) for single dose compared to placebo even after applying the Bonferroni correction (27.6 vs. 6.6; $U = 666.5$; $p = 0.002$). All other comparisons between mitragynine and placebo for Emax were not statistically significant, regardless of whether the Bonferroni correction was considered ($175 \leq Us \leq 485$; $ps > 0.05$). Mann–Whitney *U* tests also indicated that there was no detectable difference in Emax at any dose between phases when the

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Do you feel a drug effect right now?

	Single-dose phase					Multiple-dose phase (after 15 doses)				
	Mitragynine/dried kratom leaf powder (mg, PO)									
	0/0	**6.65/500**	**13.3/1000**	**26.6/2000**	**53.2/4000**	**0/0**	**6.65/500**	**13.3/1000**	**26.6/2000**	**53.2/4000**
Participants (*n*)	67	12	12	12	13	56	11	11	9	9
Emax										
Mean	6.6	2.9	18.5	11.0	27.6*	2.6	1.7	6.4	1.7	9.4
(SD)	(11.7)	(3.3)	(25.7)	(19.4)	(30.7)	(6.9)	(2.1)	(9.2)	(4.3)	(10.1)
AUC										
Mean	16.4	5.0	59.8	10.8	84.3	6.3	3.9	22.7	1.6	14.5
SD	(38.4)	(6.4)	(112.9)	(19.8)	(157.5)	(15.8)	(4.6)	(39.5)	(3.3)	(15.4)

Note: Numbers here include all participants who completed dosing for both phases. Participants completed both phases sequentially. Drug effect was queried after the first administration and after the last investigational product administration (i.e., after the 15th dose in the multiple dose phase).
Abbreviations: PO, *per os* (oral); SD, standard deviation.
*Statistically significantly different from placebo (0/0) after Bonferroni correction.

Bonferroni correction was applied ($57.5 \leq U\text{s} \leq 84.5$; $p\text{s} > 0.004$). However, Emax was greater in the second highest dose group (26.6 mg, mitragynine) for single dose compared to multiple dose when the Bonferroni correction is *not* considered (11.0 vs. 1.7; $U = 84.5$; $p = 0.03$).

3.3 | COWS and SOWS

Mann–Whitney *U* tests indicated that there was no detectable withdrawal at any dose or timepoint as measured by the COWS with or without a Bonferroni correction ($164.0 \leq U\text{s} \leq 449.5$; $p\text{s} > 0.002$ [with Bonferroni]; $p\text{s} > 0.05$ [without Bonferroni]). Similarly, Mann–Whitney *U* tests, with and without a Bonferroni correction, found no detectable withdrawal at any dose or timepoint as measured by the SOWS ($p\text{s} > 0.0007$ and $p\text{s} > 0.05$, respectively).

3.4 | Adverse Events

There were no serious adverse events (SAEs) or deaths in this study. Full listings of all AEs by incidence and frequency for the SD and MD phase are provided in Supplemental Tables 2 and 3, respectively. In the SD phase (Table S2), the proportion of participants experiencing at least one TEAE was generally higher in active treatment groups compared with placebo, although this pattern was not strictly dose-dependent. TEAEs were reported in 25.0%–76.9% of participants across active dose groups, compared with 29.9% in the pooled placebo group. The highest incidence was observed in the 53.2/4000 dose group (76.9%), while lower-dose groups demonstrated more variable rates (25.0%–41.7%). Overall, these findings suggest an increased likelihood of TEAEs with active treatment relative to placebo, particularly at higher doses.

In the MD phase (Table S3), a clearer separation between active treatment and placebo was observed. The proportion of participants reporting at least 1 TEAE ranged from 18.2% to 83.3% across active dose groups, compared with 41.7% in the placebo group. Notably, the highest TEAE incidence was again observed in the highest dose group (53.2/4000; 83.3%), while intermediate dose groups (e.g., 13.3/1000) also demonstrated relatively high rates (72.7%). Lower-dose groups showed more modest or comparable rates relative to placebo. Overall, TEAE incidence appeared to increase with dose in the MD phase, with several active treatment groups exceeding the placebo rate.

Abuse-related and withdrawal-related TEAEs are discussed below.

3.4.1 | Single-Dose Phase

Twenty-two participants reported 28 ARTEAEs in the SD phase. One episode of somnolence and 1 episode of anxiety were determined to be unrelated to investigational product; therefore, 20 participants with 26 ARTEAEs were suspected to be related to investigational product (See Table 3). All ARTEAEs in the SD phase were mild ($n = 25$, 96.2%) or

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | Abuse-related treatment-emergent Adverse Events in the Single Dose Phase.

	Mitragynine/dried kratom leaf powder (mg, PO)					
	0/0, *n* = 67	**6.65/500, *n* = 12**	**13.3/1000, *n* = 12**	**26.6/2000, *n* = 12**	**53.2/4000, *n* = 13**	**Total *N* = 116**
Participants with ARTEAE (*n*; %)	6 (9.0%)	0 (0.0%)	4 (33.3%)	3 (25.0%)	7 (53.8%)	20 (17.2%)
Participants with no ARTEAE (*n*; %)	61 (91%)	12 (100%)	8 (66.7%)	9 (75.0%)	6 (46.2%)	96 (82.8%)
Number of ARTEAEs	9	0	4	4	9	26
System organ class	**Reported frequency**					
Preferred term	**No. (%)**					
General disorders and administration site conditions	**4 (44.4%)**	**0 (0.0%)**	**2 (50.0%)**	**2 (50.0%)**	**4 (44.4%)**	**12 (46.2%)**
Feeling abnormal	1 (11.1%)	0 (0.0%)	1 (25.0%)	1 (25.0%)	0 (0.0%)	3 (11.5%)
Feeling of relaxation	1 (11.1%)	0 (0.0%)	1 (25.0%)	1 (25.0%)	2 (22.2%)	5 (19.2%)
Feeling drunk	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (22.2%)	2 (7.7%)
Energy increased	2 (22.2%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	2 (7.7%)
Nervous system disorders	**4 (44.4%)**	**0 (0.0%)**	**0 (0.0%)**	**1 (25.0%)**	**3 (33.3%)**	**8 (30.8%)**
Somnolence	4 (44.4%)	0 (0.0%)	0 (0.0%)	1 (25.0%)	3 (33.3%)	8 (30.8%)
Psychiatric disorders	**1 (11.1%)**	**0 (0.0%)**	**2 (50.0%)**	**1 (25.0%)**	**2 (22.2%)**	**6 (23.0%)**
Euphoric mood	1 (11.1%)	0 (0.0%)	1 (25.0%)	1 (25.0%)	1 (11.1%)	4 (15.4%)
Hypervigilance	0 (0.0%)	0 (0.0%)	1 (25.0%)	0 (0.0%)	1 (11.1%)	2 (7.7%)

Abbreviations: ARTEAE, abuse-related treatment-emergent adverse event; No., number; PO, *per os* (oral).

moderate ($n = 1$, 3.8%) in severity. The most commonly reported ARTEAEs in the SD phase were "somnolence" and "feeling of relaxation." The proportion of participants experiencing at least one ARTEAE was higher in active treatment groups compared with placebo and demonstrated a general increase with dose. Abuse-related TEAEs were reported in 0.0% of participants in the lowest active dose group (6.65/500), compared with 9.0% in the placebo group. Incidence increased at higher doses, with 33.3% and 25.0% of participants affected in the 13.3/1000 and 26.6/2000 groups, respectively, and reaching 53.8% in the highest dose group (53.2/4000). Overall, 17.2% of participants experienced at least one ARTEAE. These findings indicate a dose-dependent increase in abuse-related events, particularly at higher exposures, with rates exceeding placebo at all but the lowest dose.

3.4.2 | Multiple-Dose Phase

Twenty-two participants reported 154 ARTEAEs in the MD phase. One episode of somnolence was determined to be unrelated to investigational product; therefore, 22 participants with 153 ARTEAEs were suspected to be related to investigational product (see Table 4). All ARTEAEs in the MD phase were mild (153, 100.0%) in severity. The most commonly reported ARTEAEs in the MD phase were "feeling of relaxation" and "hypervigilance."

A similar but more pronounced dose-related pattern was observed in the MD phase. ARTEAEs were reported in 16.7% of participants receiving placebo. No ARTEAEs were observed in the 6.65/500 and 26.6/2000 dose groups, whereas 36.4% of participants in the 13.3/1000 group and 66.7% in the 53.2/4000 group experienced at least 1 ARTEAE. Overall, 21.2% of participants reported ARTEAEs. These results suggest a greater separation between active treatment and placebo at higher doses under repeated dosing conditions, with the highest dose group demonstrating the greatest incidence of abuse-related events.

3.4.3 | Follow-Up Phase

Only one withdrawal-related TEAE was reported during the follow-up phase; 1 participant receiving 1000 mg kratom (13.3 mg, mitragynine) reported 1 instance of mild diarrhea approximately 2 days after stopping kratom intake, which resolved in less than a day without any action taken.

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 | Abuse-related treatment-emergent adverse events in the multiple-dose phase.

	Mitragynine/dried kratom leaf powder (mg, PO)					
	0/0, *n* = 60	**6.65/500, *n* = 11**	**13.3/1000, *n* = 11**	**26.6/2000, *n* = 10**	**53.2/4000, *n* = 12**	**Total, *N* = 104**
Participants with ARTEAE (*n*; %)	10 (16.7%)	0 (0.0%)	4 (36.4%)	0 (0.0%)	8 (66.7%)	22 (21.2%)
Participants with no ARTEAE (n; %)	50 (83.3%)	11 (100%)	7 (63.6%)	10 (100%)	4 (33.3%)	82 (78.8%)
Number of ARTEAEs	53	0	31	0	69	153
System organ class	**Reported frequency**					
Preferred term	**No. (%)**					
General disorders and administration site conditions	33 (62.3%)	0 (0.0%)	14 (45.2%)	0 (0.0%)	38 (55.1%)	85 (55.6%)
Feeling abnormal	17 (32.1%)	0 (0.0%)	2 (6.5%)	0 (0.0%)	3 (4.3%)	22 (14.4%)
Feeling of relaxation	16 (30.2%)	0 (0.0%)	12 (38.7%)	0 (0.0%)	12 (17.4%)	40 (26.1%)
Feeling drunk	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	23 (33.3%)	23 (15.0%)
Nervous system disorders	12 (22.6%)	0 (0.0%)	2 (6.5%)	0 (0.0%)	7 (10.1%)	21 (13.7%)
Somnolence	12 (22.6%)	0 (0.0%)	2 (6.5%)	0 (0.0%)	7 (10.1%)	21 (13.7%)
Psychiatric disorders	8 (15.1%)	0 (0.0%)	15 (48.4%)	0 (0.0%)	24 (34.8%)	47 (30.7%)
Euphoric mood	3 (5.7%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	3 (2.0%)
Hypervigilance	3 (5.7%)	0 (0.0%)	15 (48.4%)	0 (0.0%)	15 (21.7%)	33 (21.6%)
Anxiety	1 (1.9%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	9 (13.0%)	10 (6.5%)
Irritability	1 (1.9%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	1 (0.7%)

Abbreviations: ARTEAE, abuse-related treatment-emergent adverse event; No., number; PO, *per os* (oral).

4 | Discussion

This randomized, placebo-controlled study evaluated the central nervous system effects associated with abuse potential following single and multiple daily oral doses of dried kratom leaf powder in healthy adults. The study found that kratom was well tolerated across all dose levels, with no SAEs reported. Abuse-related subjective and objective outcomes were low overall and, when present, were most prominent at the highest tested dose (53.2 mg mitragynine). Importantly, short-term administration of standardized kratom leaf powder in this sample of healthy adults did not produce clinically significant opioid withdrawal as measured by COWS/SOWS and produced modest subjective effects at the highest dose tested.

These findings are broadly consistent with preclinical and epidemiological evidence suggesting a relatively low abuse liability for mitragynine and kratom products, particularly in comparison to classical opioids. In animal models, mitragynine has not supported self-administration and does not substitute for morphine or heroin, while human survey data indicate relatively mild and infrequent withdrawal symptoms among those who abruptly discontinue moderate use; although, this is dose-related, with persons consuming higher amounts more frequently reporting more withdrawal [34]. Consistent with this prior work, the present study demonstrated that while participants endorsed some subjective effects, particularly at the highest dose, these effects were modest in magnitude and diminished with repeated exposure. Likewise, only a single mild withdrawal-related AE (diarrhea) was observed during the follow-up phase, and neither COWS nor SOWS scores differed significantly from placebo at any dose or timepoint.

Across both phases, the highest incidence of ARTEAEs consistently occurred in the highest dose group, supporting a dose–response relationship. In contrast, lower dose groups showed variable and in some cases lower incidence than placebo, particularly in the MD phase. Taken together, these findings indicate that abuse-related AEs are more likely to occur at higher doses and with repeated administration while remaining relatively infrequent at lower doses.

The dose-dependent increase in ARTEAEs, particularly in the highest dose group, may reflect pharmacodynamic activity at the μ-opioid receptor, where mitragynine acts as a partial agonist [7]. Nonetheless, the nature and severity of these AEs, most commonly somnolence, feeling of relaxation, and hypervigilance, differ from those typically observed with full μ-opioid receptor agonists such as hydrocodone or oxycodone

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[46] which often include euphoria, respiratory depression, constipation, sleepiness, etc. This is consistent with similar findings from other surveys and reviews of kratom use where consumers generally reported symptoms as mild to moderate [25, 34]. Furthermore, the diminishing subjective drug effects observed between the SD and MD phases may suggest mild tolerance development or a desensitization of effect over time, although participants did not experience significant withdrawal effects when the study concluded and discontinued use of the investigational product.

The DEQ responses in particular offer important insight. Only the highest dose group demonstrated statistically significant increases in Emax and AUC compared to placebo following single administration. After 15 days of daily administration, DEQ responses were lower across all dose groups, further supporting reduced sensitivity or subjective valuation of the kratom product with repeated exposure. Importantly, because three participants with above-average DEQ responses withdrew prior to completion of the MD phase, the interpretation of reduced subjective effects over time must be limited.

Notably, while some participants reported symptoms classified as abuse-related (e.g., hypervigilance), these occurred at low frequencies and were not associated with sustained or escalating responses. The absence of a clear withdrawal syndrome, either through clinical observation (i.e., COWS) or participant self-report (i.e., SOWS), suggests limited potential for physical dependence under these specific test conditions (i.e., short-term administration in healthy kratom-naïve individuals).

This study has several limitations. First, while the study enrolled healthy participants without recent kratom use, findings may not generalize to individuals with a history of regular kratom consumption or polysubstance use. Second, the study duration, although longer than many human abuse potential studies, may not fully capture the potential risks associated with chronic, higher dose kratom use that may be observed in some real-world settings (in one survey, participants reported use of single doses up to 15 g of kratom) [34].

5 | Conclusion

The absence of SAEs, the mild nature of most ARTEAEs, and the lack of a significant withdrawal syndrome provide support for continued investigation into kratom's utility as a dietary supplement. At the same time, the dose-related emergence of some abuse-related effects highlights the importance of dose control and product standardization in both clinical and consumer settings.

Acknowledgments

This study was supported by NP Pharma, currently doing business as Della Terra Pharmaceuticals.

Funding

This work was supported by NP Pharma d.b.a. Della Terra Pharmaceuticals.

Conflicts of Interest

This study was supported by NP Pharma, currently doing business as Della Terra Pharmaceuticals. NP Pharma, currently doing business as Della Terra Pharmaceuticals, reviewed a draft of the manuscript. NP Pharma sponsored a trip for Marion Coe to present preliminary findings at the 63rd Annual Meeting of the American College of Neuropsychopharmacology (December 8–11, 2024) in Phoenix, Arizona. Through Pinney Associates, Marion Coe, Daniel Wang, and Jack Henningfield consult to NP Pharma. In addition to NP Pharma, in the last 36 months, Pinney Associates has consulted to the American Kratom Association, Holistic Alternative Recovery Trust, and Botanic Tonics LLC (the last uncompensated) on kratom science and regulatory issues and studies. Through Pinney Associates, in support of the American Kratom Association, JH has commented on kratom pharmacology, public health, and regulatory needs for state hearings and for a judicial hearing to address questions raised by the court on these same matters. Through ALA BioPharma Consulting, Thomas Hudzik consults to NP Pharma. Through JB PharmaConsult, JB consults to NP Pharma. Through Huestis and Smith Toxicology LLC, Marilyn Huestis consults to NP Pharma. JH, through Pinney Associates, and MH are paid expert witnesses in legal cases related to kratom.

Data Availability Statement

Research data are not shared.

References

1. S. Assanangkornchai, A. Muekthong, N. Sam-angsri, and U. Pattanasattayawong, "The Use of Mitragynine speciosa ("Krathom"), an Addictive Plant, in Thailand," *Substance Use & Misuse* 42, no. 14 (2007): 2145–2157.

2. M. A. Coe, J. L. Pillitteri, M. A. Sembower, K. K. Gerlach, and J. E. Henningfield, "Kratom as a Substitute for Opioids: Results From an Online Survey," *Drug and Alcohol Dependence* 202 (2019): 24–32.

3. O. Grundmann, "Patterns of Kratom Use and Health Impact in the US—Results From an Online Survey," *Drug and Alcohol Dependence* 176 (2017): 63–70.

4. K. L. R. Jansen and C. J. Prast, "Ethnopharmacology of Kratom and the Mitragyna Alkaloids," *Journal of Ethnopharmacology* 23, no. 1 (1988): 115–119.

5. U.S. Food and Drug Administration. "FDA and Kratom." (2024): https://www.fda.gov/news-events/public-health-focus/fda-and-kratom. Accessed 29 April, 2025.

6. M. E. Reeve, S. Obeng, F. L. Oyola, et al., "The Adrenergic a2 Receptor-Mediated Discriminative-Stimulus Effects of Mitragynine, the Primary Alkaloid in Kratom (*Mitragyna speciosa*) in Rats," *FASEB Journal* 34, no. S1 (2020): 1–1.

7. A. C. Kruegel, M. M. Gassaway, A. Kapoor, et al., "Synthetic and Receptor Signaling Explorations of the Mitragyna Alkaloids: Mitragynine as an Atypical Molecular Framework for Opioid Receptor Modulators," *Journal of the American Chemical Society* 138, no. 21 (2016): 6754–6764.

8. A. C. Kruegel, R. Uprety, S. G. Grinnell, et al., "7-Hydroxymitragynine Is an Active Metabolite of Mitragynine and a Key Mediator of Its Analgesic Effects," *ACS Central Science* 5, no. 6 (2019): 992–1001.

9. J. E. Henningfield, D. W. Wang, and M. A. Huestis, "Kratom Abuse Potential 2021: An Updated Eight Factor Analysis," *Frontiers in Pharmacology* 12 (2021): 775073.

10. Q. Qu, W. Huang, D. Aydin, et al., "Insights into Distinct Signaling Profiles of the μOR Activated by Diverse Agonists," *Nature Chemical Biology* 19, no. 4 (2023): 423–430.

11. C. R. McCurdy, A. Sharma, K. E. Smith, et al., "An Update on the Clinical Pharmacology of Kratom: Uses, Abuse Potential, and Future

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Considerations," *Expert Review of Clinical Pharmacology* 17, no. 2 (2024): 131–142.

12. S. Chakraborty, J. F. DiBerto, A. Faouzi, et al., "A Novel Mitragynine Analog With Low-Efficacy Mu Opioid Receptor Agonism Displays Antinociception With Attenuated Adverse Effects," *Journal of Medicinal Chemistry* 64, no. 18 (2021): 13873–13892.

13. J. D. Zuarth Gonzalez, M. Mottinelli, C. R. McCurdy, G. de Lartigue, L. R. McMahon, and J. L. Wilkerson, "Mitragynine and Morphine Produce Dose-Dependent Bimodal Action on Food but Not Water Intake in Rats," *American Journal of Physiology. Regulatory, Integrative and Comparative Physiology* 327, no. 6 (2024): R568–r579.

14. D. A. Todd, J. J. Kellogg, E. D. Wallace, et al., "Chemical Composition and Biological Effects of Kratom (*Mitragyna speciosa*): In Vitro Studies With Implications for Efficacy and Drug Interactions," *Scientific Reports* 10, no. 1 (2020): 19158.

15. P. K. Manwill, L. Flores-Bocanegra, M. Khin, et al., "Kratom (*Mitragyna speciosa*) Validation: Quantitative Analysis of Indole and Oxindole Alkaloids Reveals Chemotypes of Plants and Products," *Planta Medica* 88, no. 09/10 (2022): 838–857.

16. N. Sengnon, P. Vonghirundecha, W. Chaichan, et al., "Seasonal and Geographic Variation in Alkaloid Content of Kratom (*Mitragyna speciosa* (Korth.) Havil.) from Thailand," *Plants* 12, no. 4 (2023): 949.

17. O. Grundmann, A. Garcia-Romeu, C. R. McCurdy, et al., "Not All Kratom Is Equal: The Important Distinction Between Native Leaf and Extract Products," *Addiction* 119, no. 1 (2024): 202–203.

18. J. Heywood, S. Smallets, and D. Paustenbach, "Beneficial and Adverse Health Effects of Kratom (*Mitragyna speciosa*): A Critical Review of the Literature," *Food and Chemical Toxicology* 192 (2024): 114913.

19. S. H. Kamble, E. C. Berthold, S. R. R. Kanumuri, et al., "Metabolism of Speciociliatine, an Overlooked Kratom Alkaloid for Its Potential Pharmacological Effects," *AAPS Journal* 24, no. 5 (2022): 86.

20. J. D. Zuarth Gonzalez, A. K. Ragsdale, S. Mukhopadhyay, et al., "Mitragynine and 7-Hydroxymitragynine: Bidirectional Effects on Breathing in Rats," *bioRxiv: The Preprint Server for Biology* (2025): 2025.2005.2016.654392.

21. J. E. Henningfield, J. V. Rodricks, A. M. Magnuson, and M. A. Huestis, "Respiratory Effects of Oral Mitragynine and Oxycodone in a Rodent Model," *Psychopharmacology* 239, no. 12 (2022): 3793–3804.

22. S. E. Hemby, S. McIntosh, F. Leon, S. J. Cutler, and C. R. McCurdy, "Abuse Liability and Therapeutic Potential of the *Mitragyna speciosa* (kratom) Alkaloids Mitragynine and 7-Hydroxymitragynine," *Addiction Biology* 24, no. 5 (2019): 874–885.

23. K. Yue, T. A. Kopajtic, and J. L. Katz, "Abuse Liability of Mitragynine Assessed With a Self-Administration Procedure in Rats," *Psychopharmacology* 235, no. 10 (2018): 2823–2829.

24. K. E. Smith, K. E. Dunn, J. M. Rogers, et al., "Kratom Use as More Than a "Self-Treatment"," *American Journal of Drug and Alcohol Abuse* 48, no. 6 (2022): 684–694.

25. K. E. Smith, L. V. Panlilio, J. D. Feldman, et al., "Ecological Momentary Assessment of Self-Reported Kratom Use, Effects, and Motivations Among US Adults," *JAMA Network Open* 7, no. 1 (2024): e2353401–e2353401.

26. O. Grundmann, M. Green, E. Berthold, S. L. Yoon, and D. Ray, "Prevalence and Use Patterns of Kratom (Mitragyna speciosa Korth.) in a US Nationally Representative Sample," *Journal of Psychoactive Drugs* (2025): 1–9.

27. M. Zhang, A. Lyndon, B. Pearson, C. McCurdy, A. Sharma, and J. Chen, "Kratom: Botanical Insights and Cultivation Practices for a Conspicuous Medicinal Tree Species: ENH1389/EP653, 7/2025," *EDIS* 2025, no. 4 (2025), https://doi.org/10.32473/edis-ep653-2025.

28. P. N. Brown, J. A. Lund, and S. J. Murch, "A Botanical, Phytochemical and Ethnomedicinal Review of the Genus Mitragyna korth: Implications for Products Sold as Kratom," *Journal of Ethnopharmacology* 202 (2017): 302–325.

29. B. P. Giroir "2018 Letter From the Assistant Secretary of Health to the Administrator of the Drug Enforcement Administration to Rescind Previous Support to Permanently Place Mitragynine and 7-Hydroxymitragynine in Schedule I of the Controlled Substances Act," 2018 https://images.go02.informamarkets.com/Web/Informa02/%7b548e6d56-2ea4-4da4-9404-0348b56e9a88%7d_dhillon-8.16.2018-response-letter-from-ash-radm-giroir.pdf.

30. I. S. Johari, N. Harun, Z. M. Sofian, and M. Shoaib, "Pentylenetetrazol-Like Stimulus Is not Produced Following Naloxone-Precipitated Mitragynine Withdrawal in Rats," *Psychopharmacology* 238, no. 11 (2021): 3183–3191.

31. A. Khazaeli, J. M. Jerry, and M. Vazirian, "Treatment of Kratom Withdrawal and Addiction With Buprenorphine," *Journal of Addiction Medicine* 12, no. 6 (2018): 493–495.

32. C. N. Stanciu, A. Gs, A. Saeed, and T. Penders, "Kratom Withdrawal: A Systematic Review With Case Series," *Journal of Psychoactive Drugs* 51, no. 1 (2019): 12–18.

33. A. Garcia-Romeu, D. J. Cox, K. E. Smith, K. E. Dunn, and R. R. Griffiths, "Kratom (*Mitragyna speciosa*): User Demographics, Use Patterns, and Implications for the Opioid Epidemic," *Drug and Alcohol Dependence* 208 (2020): 107849.

34. J. M. Rogers, S. T. Weiss, D. H. Epstein, O. Grundmann, K. Hill, and K. E. Smith, "Kratom Addiction Per DSM-5 SUD Criteria, and Kratom Physical Dependence: Insights From Dosing Amount Versus Frequency," *Drug and Alcohol Dependence* 260 (2024): 111329.

35. M. T. Swogger, H. Elaine, E. Fire, et al., "Experiences of Kratom Users: A Qualitative Analysis," *Journal of Psychoactive Drugs* 47, no. 5 (2015): 360–367.

36. O. Grundmann, J. K. Babin, J. E. Henningfield, et al., "Kratom Use in the United States: A Diverse and Complex Profile," *Addiction* 116, no. 1 (2021): 202–203.

37. M. T. Swogger and Z. Walsh, "Kratom Use and Mental Health: A Systematic Review," *Drug and Alcohol Dependence* 183 (2018): 134–140.

38. M. M. Sweeney, D. C. Weaver, K. B. Vincent, A. M. Arria, and R. R. Griffiths, "Prevalence and Correlates of Caffeine Use Disorder Symptoms Among a United States Sample," *Journal of Caffeine and Adenosine Research* 10, no. 1 (2020): 4–11.

39. M. Bodur, S. Kaya, M. Ilhan-Esgin, F. P. Çakiroğlu, and A. Özçelik, "The Caffeine Dilemma: Unraveling the Intricate Relationship Between Caffeine Use Disorder, Caffeine Withdrawal Symptoms and Mental Well-Being in Adults," *Public Health Nutrition* 27, no. 1 (2024): e57.

40. F. Abdoli, M. Davoudi, F. Momeni, et al., "Estimate the Prevalence of Daily Caffeine Consumption, Caffeine Use Disorder, Caffeine Withdrawal and Perceived Harm in Iran: A Cross-Sectional Study," *Scientific Reports* 14, no. 1 (2024): 7644.

41. J. E. Henningfield, M. C. Chawarski, A. Garcia-Romeu, et al., "Kratom Withdrawal: Discussions and Conclusions of a Scientific Expert Forum," *Drug and Alcohol Dependence Reports* 7 (2023): 100142.

42. M. A. Huestis, M. A. Brett, J. Bothmer, and R. Atallah, "Human Mitragynine and 7-Hydroxymitragynine Pharmacokinetics After Single and Multiple Daily Doses of Oral Encapsulated Dried Kratom Leaf Powder," *Molecules* 29, no. 5 (2024): 984.

43. M. A. Huestis, J. Bothmer, T. Hudzik, J. E. Henningfield, and S. Swift, "Safety and Tolerability of Single and Multiple Daily Oral Doses of Dried Kratom Leaf Powder in a Randomized Trial in Healthy Volunteers," *Therapeutic Drug Monitoring* (2026).

44. S. Galati, "Identifying Relevant Adverse Events of Interest and Recommendation for Analysis and Presentation of Data in the NDA

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Submission," [Presented at Cross-Company Abuse Liability Council (CCALC) Meeting] September 27, 2023; Rockville, MD.

45. T. R. Kosten and L. E. Baxter, "Review Article: Effective Management of Opioid Withdrawal Symptoms: A Gateway to Opioid Dependence Treatment," *American Journal on Addictions* 28, no. 2 (2019): 55–62.

46. S. L. Walsh, P. A. Nuzzo, M. R. Lofwall, and J. R. Holtman, "The Relative Abuse Liability of Oral Oxycodone, Hydrocodone and Hydromorphone Assessed in Prescription Opioid Abusers," *Drug and Alcohol Dependence* 98, no. 3 (2008): 191–202.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Recommended list of abuse-related AEs to assess abuse potential as presented at the Cross-Company Abuse Liability Counsel (CCALC; Galati, S. (2023). Identifying Relevant Adverse Events of Interest and Recommendation for Analysis and Presentation of Data in the NDA Submission. Cross-Company Abuse Liability Council (CCALC) Meeting. Rockville, MD). **Table S2:** Treatment-emergent adverse events in the single-dose phase. **Table S3:** Treatment-emergent adverse events in the multiple-dose phase.

19427611, 2026, 7, Downloaded from https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/10.1002/dta.70089, Wiley Online Library on [09/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License